

CHEMISTRY

Q1 Which of the following given atomic number have the lowest melting point

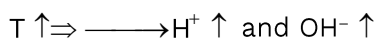
- (1) 14 (2) 6 (3) 82 (4) 50

Q2 What will be effect on pH of water when it is heated

- (1) Increase (2) Decrease
(3) Remains same (4) pH first increases then decreases

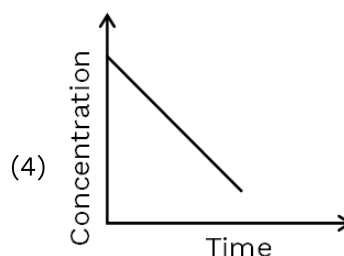
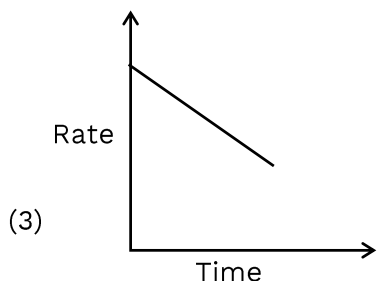
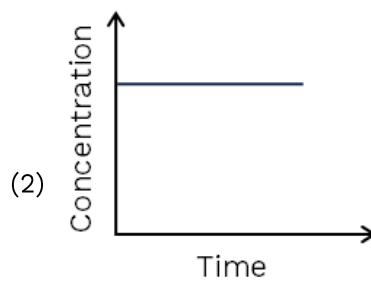
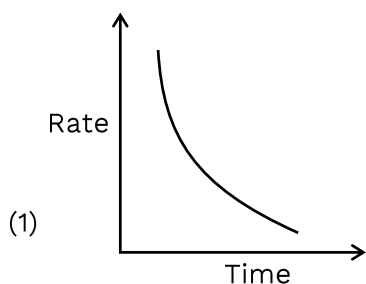
Sol. $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$

$$\Delta H = +ve$$



$$\text{pH} = \frac{k_w}{2} = \frac{-\log k_w}{2} \downarrow k_w = [\text{H}^+][\text{OH}^-] \uparrow$$

Q3 Which of the following is correct for zero order reaction



Ans. (4)

Sol. $\text{A} \xrightarrow{k} \text{P}$

$$(n=0)$$

$$r = k$$

$$a_t = a_0 - kt$$

Q4 **List-I**

(Alloys)

(P) Brass

(Q) Bronze

(R) Stainless Steel

(S) Silver Coin

(1) P→2; Q→4; R→1; S→3

(3) P→2; Q→3; R→1; S→4

List-II

(Metals)

(1) Fe, Cr, Ni

(2) Cu, Zn

(3) Ag, Cu, Zn, Ni

(4) Cu, Sn

(2) P→3; Q→4; R→1; S→2

(4) P→1; Q→2; R→3; S→4

Ans. (1)

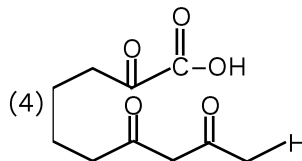
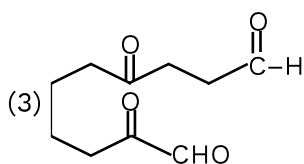
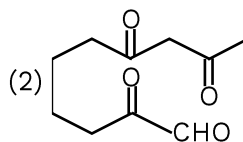
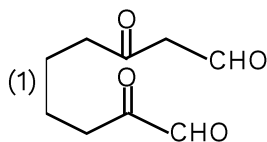
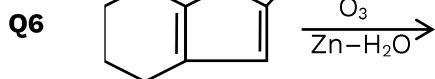
Sol. Factual base

Q5 α -helix protein and β -pleated sheet structure protein belongs to which type of protein ?

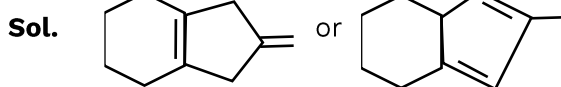
- (1) Primary (2) Secondary (3) Tertiary (4) Quarternary

Ans. (2)

Sol. Factual



Ans. (2)



Q7 Identify the number of complex in which central metal ion has d^4 configuration

$[\text{FeO}_4]^{2-}$, $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{NiF}_6]^{2-}$, $[\text{NiCl}_4]^{2-}$, $\text{Cr}_2(\text{CH}_3\text{COO})_4(\text{H}_2\text{O}_2)_2$

Ans. 2

Sol. $\text{Fe}^{+6} = 3d^2$

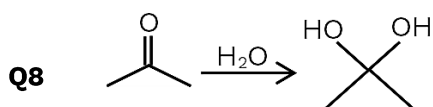
$\text{Mn}^{+3} = 3d^4$

$\text{Fe}^{+3} = 3d^5$

$\text{Ni}^{+4} = 3d^6$

$\text{Ni}^{+2} = 3d^8$

$\text{Cr}^{+2} = 3d^4$



Given below are two statements :

Statement (I):  (r = 22400) rate due to less steric

Statement (II): $\text{CCl}_3\text{-CHO}$ (r = 2000) rate due to $-\text{I}$ group of halogen

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement I and Statement II are incorrect
 (2) Statement I is correct but Statement II is incorrect
 (3) Both Statement I and Statement II are correct
 (4) Statement I is incorrect but Statement II is correct

Ans. (3)

Sol. Factual

Q9 Given below are two statements :

Statement (I): Phenol & alcohol melting point increases with increase in carbon atoms

Statement (II): Phenol & alcohol has higher melting point than ether and haloalkanes

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

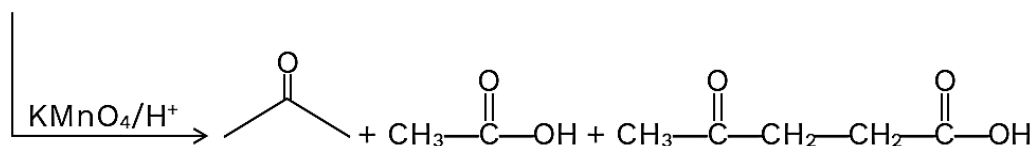
(4) Statement I is incorrect but Statement II is correct

Ans. (3)

Sol. Statement-1: Weight $\uparrow \rightarrow$ Molecular weight $\uparrow$ Melting point $\uparrow$

Statement-2: due to hydrogen bonding

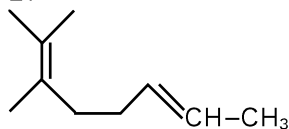
Q10 Compound (X) $\xrightarrow{\text{Hydrogenation}}$ 2 mole H_2 Consumed



Number of σ bonds in compound (X) ?

Ans. 27

Sol.



$X = 27$

Q11 Standard electrode potential for a half cell are mention below

$E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34\text{V}$ $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = 0.76\text{V}$ $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80\text{V}$ $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37\text{V}$

Which of the following cells gives the most negative value of ΔG°

(1) $\text{Zn}/\text{Zn}^{2+} (1\text{M}) \parallel \text{Mg}^{2+} (1\text{M})/\text{Mg}$

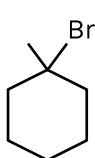
(2) $\text{Cu}/\text{Cu}^{2+} (1\text{M}) \parallel \text{Ag}^+ (1\text{M})/\text{Ag}$

(3) $\text{Zn}/\text{Zn}^{2+} (1\text{M}) \parallel \text{Ag}^+ (1\text{M})/\text{Ag}$

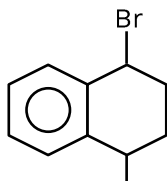
(4) $\text{Ag}/\text{Ag}^+ (1\text{M}) \parallel \text{Mg}^{2+} (1\text{M})/\text{Mg}$

Sol. $E^\circ = E^\circ_{\text{C}} - E^\circ_{\text{A}}$

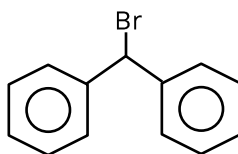
Q12 Descending order of selective state of solvolysis of following compound



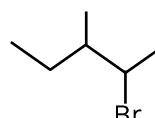
(I)



(II)



(III)



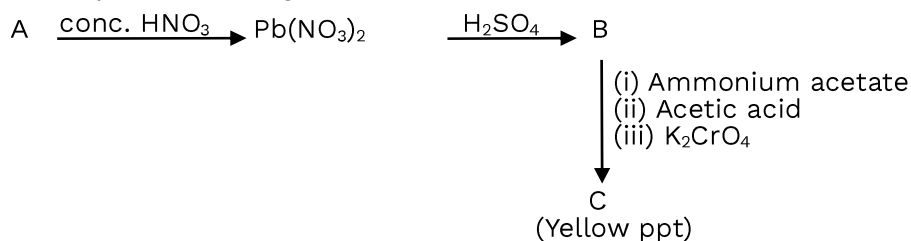
(IV)

(1) $\text{III} < \text{II} < \text{I} < \text{IV}$

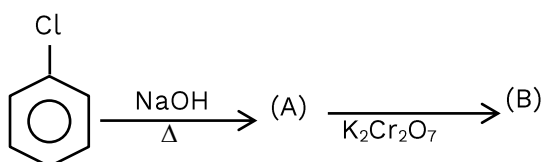
(2) $\text{II} < \text{III} < \text{I} < \text{IV}$

(3) $\text{III} < \text{I} < \text{II} < \text{IV}$

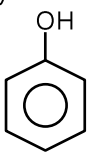
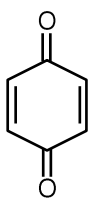
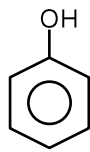
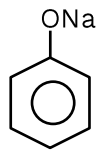
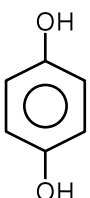
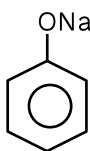
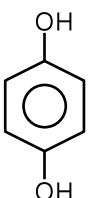
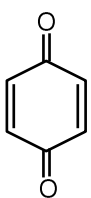
(4) $\text{III} > \text{II} > \text{I} > \text{IV}$

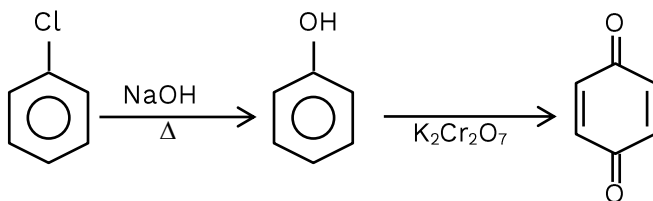
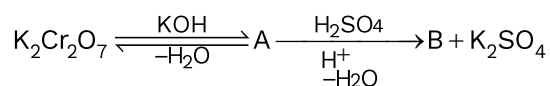
Ans. (4)**Sol.** SN_1 – Stable carbocation**Q13** Identify A,B,C in the given reaction**Sol.** Pbs dissolved in conc. HNO_3

So

 $A = \text{PbS} \downarrow$ $B = \text{PbSO}_4 \downarrow$ $C = \text{PbCrO}_4 \downarrow$ **Q14**

Identify A & B

- (1)  ; 
- (2)  ; 
- (3)  ; 
- (4)  ; 

Ans. (1)**Sol.****Q15** Identify A & B in the given reaction

- (1) K_2CrO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ (2) CrO_3 , K_2CrO_4 (3) K_2CrO_4 , $\text{Cr}_2(\text{SO}_4)_3$ (4) $\text{Cr}(\text{OH})_3$, $\text{K}_2\text{Cr}_2\text{O}_7$

Ans. (1)

Sol. Dichromate $\xrightarrow{\text{OH}^-}$ Chromate
 $(\text{K}_2\text{Cr}_2\text{O}_7)$ (K_2CrO_4)

Q16 Given below are two statements :

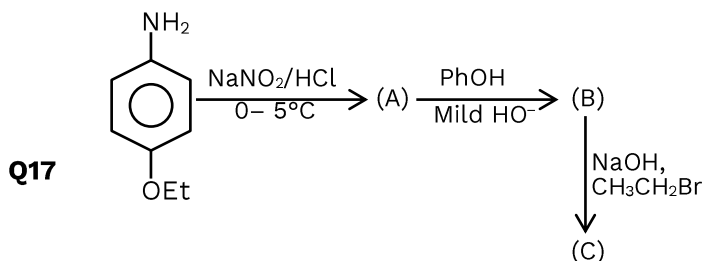
Statement (I): For a given shell, the total number of allowed orbitals is given by n^2 .

Statement (II): For any subshell the spatial orientation of the orbital is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the most appropriate answer from the options given below :

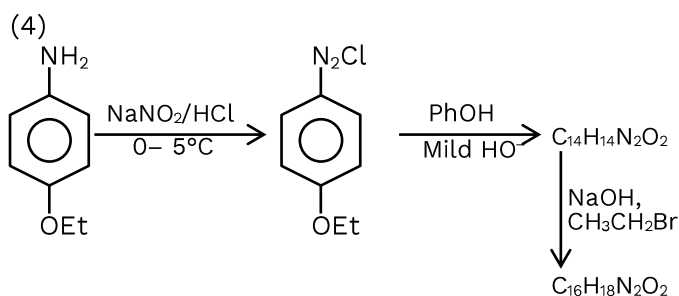
- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Both Statement I and Statement II are correct
- (4) Statement I is incorrect but Statement II is correct

Sol. factual based



Total sp^3 hybridized carbon atoms in product (C)

Ans.



Sol.

Q18 When a non volatile solute (A) is added to the solvent, the vapour pressure of the solvent decreases by 10mm of Hg. The mole fraction of the solute in the solution 0.2. If second solute (B) is added to the same solution and vapour pressure of solution decrease by 20mm Hg. Calculate mole fraction of second solute in the final solution

- (1) 0.3
- (2) 0.4
- (3) 0.5
- (4) 0.6

Sol.

$$\frac{\Delta p}{p^\circ} = X_{\text{solut.}}$$

$$(1) \quad \frac{10}{p^\circ} = 0.2 \Rightarrow p^\circ = \frac{10}{0.2} = 50 \text{ mm}$$

$$n_T = 1 \Rightarrow n_A = 0.2$$

$$(2) \quad n_{\text{Solvent}} = 0.8$$

$$\frac{30}{50} = \frac{0.2 + n_B}{1 + n_B} \Rightarrow 0.2 + n_B = 0.6 + 0.6n_B$$

$$0.4n_B = 0.4$$

$$X_B = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

$$n_B=1$$

Q19 By using relation

$$\Delta G = \Delta H - T\Delta S$$

Which of the following is incorrect for spontaneous reaction at a given temperature

(1) $\Delta H > 0, \Delta S > 0$

(2) $\Delta H > 0, \Delta S < 0$

(3) $\Delta H < 0, \Delta S > 0$

(4) $\Delta H < 0, \Delta S < 0$

Sol. $\Delta H = -ve$

Or $\Delta S = +ve$

Q20 0.01 mole of an organic compound (Hydrocarbon) gives 1.76 gm CO_2 and 0.9 gm H_2O on complete combustion. Find out chemical formula of compound.

(1) C_3H_8

(2) C_4H_{10}

(3) C_5H_{12}

(4) C_6H_{14}

Sol. $\text{C}_x\text{H}_y + \left(x + \frac{y}{4}\right)\text{O}_2 \rightarrow x\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$

0.01 mole]

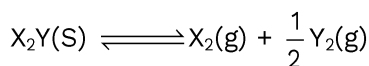
$$0.01x \quad \frac{0.01}{2}y$$

$$= \frac{1.76}{44} = \left(\frac{0.9}{18}\right)$$

$$x = \frac{176}{44} = 4$$

$$y = \frac{180}{18} = 10$$

Q21 Consider the given following reaction



If α is the degree of the dissociation.

Calculate K_p in terms of P total pressure.

(1) $K_p = \sqrt{\frac{2P}{3}}$

(2) $K_p = \frac{\sqrt{2P}}{3}$

(3) $K_p = \frac{2P^{3/2}}{3^{3/2}}$

(4) $K_p = \frac{2P^{3/2}}{3}$

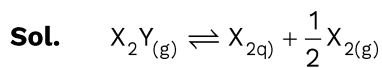
Q22 Consider a binary solution of two volatile liquid components 1 and 2. x_1 and y_1 are the small fraction of component 1 in liquid and vapour pressure respectively. The slope and intercept of linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given as

(1) $\frac{P_1^0}{P_2^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

(2) $\frac{P_2^0}{P_1^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

(3) $\frac{P_2^0}{P_1^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

(4) $\frac{P_1^0}{P_2^0}, \frac{P_1^0 - P_2^0}{P_2^0}$



$$t = 0 \quad P^\circ$$

$$t_{eq} \quad P^\circ(1-\alpha) \quad P^\circ\alpha + \frac{P^\circ\alpha}{2}$$

$$P = P^\circ \left(1 + \frac{\alpha}{2} \right) \quad (1)$$

$$K_p = \frac{P^\circ\alpha + \left(\frac{P^\circ\alpha}{2} \right)^{1/2}}{P^\beta(1-\alpha)} = \frac{\alpha}{(1-\alpha)} \times \left[\frac{P}{\left(1 + \frac{\alpha}{2} \right)^2} \times \frac{\alpha}{2} \right]^{1/2}$$

$$= \alpha \left[\frac{P\alpha}{2} \right]^{1/2}$$

$$(K_p)^2 = \alpha^2 \times \frac{P\alpha}{2} \Rightarrow \alpha^3 = \frac{2(K_p)^2}{P} \Rightarrow \left(\frac{2(K_p)^2}{P} \right)^3$$